

Supplemental Appendix

The Inequality Multiplier: Market Inelasticity and the Persistence of Wealth Inequality

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1. Inelastic Investors

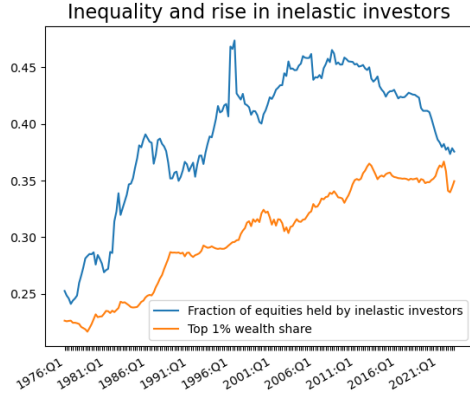
As shown in Figure 1A, the share of wealth held by the top 1% in the United States has risen from below 25% in 1976 to 35% in 2021 (Saez and Zucman (2016)). Over the same period, the fraction of the total U.S. equity market held by institutional investment funds (such as Mutual Funds, ETFs, and Insurance funds) has risen from 25% to 40%. These investors are characterized by a relatively low price elasticity of demand for equity (or inelasticity), which leads to a relatively more stable equity share compared to their elastic counterparts (such as hedge funds and households) over the business cycle (Figure 1B). The growing presence of inelastic investors makes the aggregate demand curve in equity markets more inelastic over time, implying that changes in household or institutional equity demand move asset prices.

To make this point more clearly, Figure 2 plots the cross-correlation between the change in top 10% (or top 1%) wealth share against a measure of the change in the macro-market elasticity¹ over time in the U.S. Each point represents a quarter between 1990 and 2021. Both plots demonstrate that over the past three decades, as market elasticity has reduced, wealth inequality has increased. The purpose of this paper is to understand to what extent can inelastic markets explain the increase in wealth inequality.

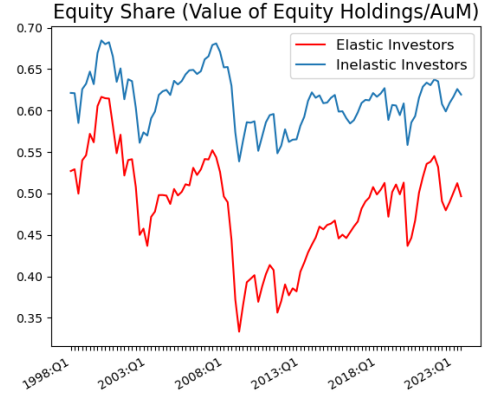
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¹The measure of elasticity is derived from Gabaix and Koijen (2021) and constructed by dividing the value of equity holdings of inelastic investors (PQ) by the value of their debt holdings (B). We motivate this measure of elasticity further in our model.



A. Rise in inequality and inelastic investors



B. Portfolio behaviour of investors

FIGURE 1. Panel A shows the wealth share of the top 1% households in the U.S. The wealth share has risen from around 23% up to 35%. Panel B shows the holdings of inelastic investors as a share of the total market cap of equities. This has risen from 25% to over 35%.

Source: Authors' calculation from Flow of Funds Data and [Saez and Zucman \(2016\)](#). Inelastic Investors include Mutual Funds, Closed-end funds, ETFs, Government Retirement Funds, Insurance funds.

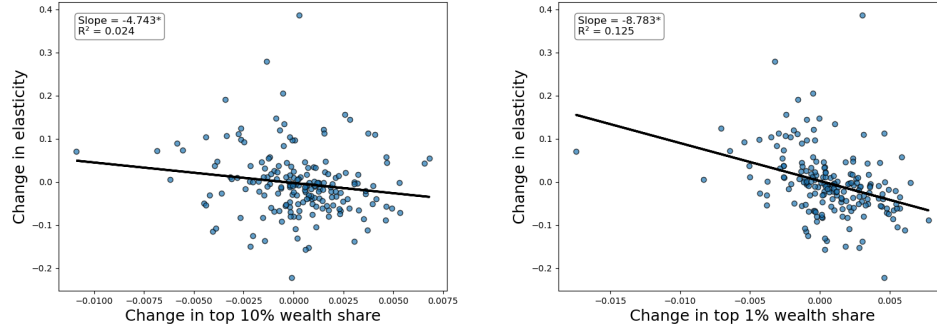


FIGURE 2. Panel A shows a significant negative correlation between the change in top 10% wealth share between 1990 and 2021 and change in market elasticity. Panel B shows a larger negative correlation between the change in top 1% wealth share and change in market elasticity.

Note: Inelastic Investors include Mutual Funds, Closed-end funds, ETFs, Government Retirement Funds, Insurance funds. In Appendix 1, we plot the actual time series for the share of equities held by inelastic investors, and how inelastic investors are characterised by more stable equity portfolio shares relative to their elastic counterparts.

Source: Quarterly top 10% and top 1% wealth shares from [Saez and Zucman \(2016\)](#) and PQ/B from flow of funds data (between 1990 and 2021).

2. Micro-foundations for Market Inelasticity

Institutional constraints and mandates contribute significantly, as shown by [Chinco and Sammon \(2024\)](#). For example, mutual funds, which own 22% of U.S. equities, and ETFs, which own 6.4%, are routinely subject to explicit or implicit investment mandates which reduce their ability to

exercise discretion in day-to-day portfolio allocation. [He and Xiong \(2013\)](#) use an agency-based model to argue that tight investment mandates are justified for most fund managers except those with exceptional talents, using this to argue for why we empirically observe the use of stringent investment mandates.

Other micro-foundations include the transactional costs of moving wealth between liquid and illiquid assets in [Kaplan et al. \(2018\)](#), which reduces the ability of households to seamlessly adjust portfolio shares invested in different asset classes. Slow-moving capital in [Duffie \(2010\)](#) (due to institutional impediments to trade such as search costs, time to raise capital or investor inattention) can generate sharp price reactions in the short- to medium-run. Preferred habitat investors in [Vayanos and Vila \(2021\)](#) and [Chen et al. \(2012\)](#) create limits to arbitrage, thereby implying that asset markets are not fully elastic. More generally, behavioral frictions such as inattention may also help explain limited ability of investors to absorb large flows into and out of asset markets.

[Haddad and Muir \(2021\)](#) use time-varying risk aversion of financial intermediaries along with costly direct investment in asset markets by households to show that the elasticity of household asset demand to asset prices is decreasing in the risk aversion of financial intermediaries as well as the cost of direct investment, generating inelastic demand curves. In a similar vein, [He and Krishnamurthy \(2018\)](#) review approaches under which institutional frictions such as equity capital constraints ([He and Krishnamurthy \(2013\)](#)), endogenous borrowing constraints ([Brunnermeier and Sannikov \(2014\)](#)), value-at-risk regulations ([Adrian and Shin \(2010\)](#)) and leverage regulations ([Du et al. \(2023\)](#)) create downward sloping demand curves for assets, specifically for large and well-regulated financial intermediaries such as commercial banks, investment banks and hedge funds.

These results carry over to other financial institutions such as mutual funds and asset managers, focusing in turn on skills of mutual fund managers ([Kacperczyk et al. \(2016\)](#)), limits to information acquisition ([Gârleanu and Pedersen \(2018\)](#)) and benchmarking considerations ([Basak and Pavlova \(2013\)](#), [Pavlova and Sikorskaya \(2023\)](#)), all of which result in the reduced ability (or appetite) of asset managers to adjust portfolios frequently and rapidly, thereby generating inelasticity in asset markets.

A natural question is why macro-arbitrageurs have not emerged to take advantage of all the above frictions, thereby offsetting the inelasticity generated due to these frictions. [Gabaix and Koijen \(2021\)](#) document limited size and influence of arbitrageurs, showing that they account for less than 5% of the overall equity market holdings. They also note that hedge funds seem to respond to crisis episodes by selling assets rather than buying, the transfer of equity risk across different types of investors is minimal, and households' equity share has shown little variation over time. Finally, [Haddad et al. \(2024\)](#) highlight that competitive strategic response of investors to other investors reducing their elasticity of demand is limited, further contributing to market inelasticity. These suggest that forces which might offset market inelasticity are limited.

Taken together, there is ample evidence for the inelasticity of equity markets, which we take as given. For ease of modelling, we introduce inelasticity through investment mandates. However, any of the above features could be incorporated to generate qualitatively similar results.

3. Additional Qualitative Results

3.1. Price Decomposition and Downward-sloping Demand

As equity prices depend upon flows (or borrowing), which in turn depend upon endowments, we can derive an expression which makes the fundamental drivers of prices explicit. Using the derived value for borrowing from Equation 18:

PROPOSITION 1. *The price of equity depends on dividends, the income distribution, equity shares outstanding, and the financial intermediary constraint:*

$$(1) \quad p_0 = \frac{\beta}{1 + \beta} \left[\frac{D_1}{(1 + r_{f,-1})^2} \right] + \frac{1}{1 + \beta} \left[\frac{1}{Q} \frac{\theta}{1 - \theta} \left\{ \frac{e_1^P - \frac{c_1^P}{c_0^P} e_0^P}{1 + r_{f,0}} \right\} \right].$$

PROOF. Note that the proof assumes a log utility function. Start with the price of equity as a function of borrowing by non-wealthy households, Equation 17:

$$p_0 = p_{-1} + \frac{\theta}{1 - \theta} \frac{B_0^P}{Q_0}$$

Use budget constraint of the non-wealthy, Equation 10 to write $B_0^P = c_0^P - e_0^P$. Combining $t = 0$ and $t = 1$ budget constraint yields the following intertemporal budget constraint:

$$c_0^P + \frac{c_1^P}{1 + r_{f,0}} + \frac{(1 + r_{f,-1})^2 B_{-1}}{1 + r_{f,0}} = e_0^P + \frac{e_1^P}{1 + r_{f,0}}$$

Under log utility, the non-wealthy Euler equation yields that $c_1^P = \beta(1 + r_{f,0})c_0^P$. Substituting and cross-multiplying:

$$c_0^P = \frac{1}{1 + \beta} \left(e_0^P + \frac{e_1^P - (1 + r_{f,-1})^2 B_{-1}}{1 + r_{f,0}} \right)$$

Substituting into the pricing equation yields:

$$p_0 = p_{-1} + \frac{\theta}{1 - \theta} \frac{1}{Q_0} \left(\frac{e_1^P}{(1 + \beta)(1 + r_{f,0})} - \frac{\beta}{1 + \beta} e_0^P \right) - \frac{\theta}{1 - \theta} \frac{1}{Q_0} \left(\frac{(1 + r_{f,-1})^2 B_{-1}}{(1 + \beta)(1 + r_{f,0})} \right)$$

Substituting $p_{-1} = \frac{D_1}{(1 + r_{f,-1})^2}$, $\frac{B_{-1}}{1 - \theta} = W_{-1} = \frac{p_{-1} Q_{-1}}{\theta}$ from Equations 2 and 4, we obtain:

$$\begin{aligned} p_0 &= p_{-1} \left(1 - \frac{(1 + r_{f,-1})^2}{(1 + \beta)(1 + r_{f,0})} \right) + \frac{\theta}{1 - \theta} \frac{1}{Q_0} \left(\frac{e_1^P}{(1 + \beta)(1 + r_{f,0})} - \frac{\beta}{1 + \beta} e_0^P \right) \\ &= \frac{D_1}{(1 + r_{f,-1})^2} \left(1 - \frac{(1 + r_{f,-1})^2}{(1 + \beta)(1 + r_{f,0})} \right) + \frac{\theta}{1 - \theta} \frac{1}{Q_0} \frac{1}{1 + \beta} \left(\frac{e_1^P}{1 + r_{f,0}} - \beta e_0^P \right) \end{aligned}$$

Invoking the expectations hypothesis, $(1 + r_{f,-1})^2 = (1 + r_{f,-1 \rightarrow 0})(1 + r_{f,0})$. However, $r_{f,-1 \rightarrow 0} = 0$ in equilibrium as no agents can buy or sell risk-free contracts from $t = -1$ to $t = 0$. Therefore, $(1 + r_{f,-1})^2 = (1 + r_{f,0})$. Then, substituting and rearranging, we obtain the final outcome:

$$p_0 = \frac{\beta}{1 - \beta} \frac{D_1}{(1 + r_{f,-1})^2} + \frac{1}{1 + \beta} \frac{\theta}{1 - \theta} \frac{1}{Q_0} \left(\frac{e_1^P}{1 + r_{f,0}} - \beta e_0^P \right)$$

□

This expression illustrates the critical role of income distribution in determining equity prices. Specifically, the heterogeneity in agents' endowments and consumption decisions directly impacts the pricing mechanism. This result deviates from [Grossman and Shiller \(1981\)](#), who argue that heterogeneity has no effect on equity prices. By incorporating heterogeneity into the model, we

show that the distribution of income plays a non-trivial role in shaping market outcomes.

Unlike models such as [Kiyotaki and Moore \(1997\)](#) where one type of households is incentivised to borrow due to differing discount rates or preferences, we assume the same level of patience and preferences across both households. Instead, we establish conditions on endowments of non-wealthy households which lead them to borrow a positive amount of debt. Quite intuitively, this turns out to be the following condition:

$$e_1^P - \frac{c_1^P}{c_0^P} e_0^P > 0 \Leftrightarrow \frac{e_1^P}{e_0^P} > \beta(1 + r_{f,0})$$

where we substitute for $\frac{c_1^P}{c_0^P}$ using the Euler Equation for non-wealthy households, $\beta \frac{u'[\frac{c_1^P}{c_0^P}]}{u'[\frac{c_0^P}{c_0^P}]}(1 + r_{f,0}) = 1$. Therefore, whenever the growth in the level of absolute endowments of the non-wealthy is greater than the growth rate of consumption (i.e. endowments are enough to cover repayment of debt), the non-wealthy borrow a positive amount. This is in line with reality — while the income *share* of the bottom 50% has reduced, their income *levels* have increased over time, allowing them to sustain more debt.

The proposition also highlights the influence of market structure. A lower quantity of equity shares outstanding leads to a higher price impact for a given level of flow, while a higher inelasticity parameter, θ , amplifies price movements further.

In [Fig. 3](#), we compare the demand curve implied by our model with a benchmark elastic demand curve derived from [Lucas \(1978\)](#). Under perfectly elastic demand, the curve is flat, indicating that even small price changes not justified by fundamentals lead to substantial adjustments in portfolio holdings. However, the demand curve from our model is downward-sloping, reflecting the inelastic nature of the market. This demonstrates that price adjustments are required to absorb flows, even in the absence of fundamental changes.

3.2. Elasticity in Three-Period Model

Note that the first derivative of equity price with respect to quantity is negative, $\frac{\partial p_0}{\partial Q_0} < 0$. This demonstrates that inelastic markets inherently generate downward-sloping demand curves, as

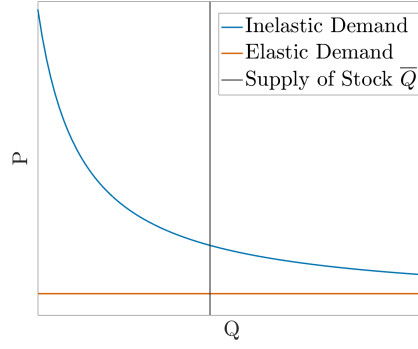


FIGURE 3. Inelastic demand curve ($\xi = 0.2$) versus elastic demand curve.

price movements are required to balance supply and demand. In such a system, what determines the elasticity of demand for equities? Consider the following proposition:

PROPOSITION 2. *The (signed) price elasticity of demand for equities ξ is determined by the (fixed) fraction of portfolio held by the financial intermediary in equities, θ :*

$$(2) \quad \xi = 1 - \theta.$$

PROOF. Combining Equations 6 and 8, quantity of equity demanded at $t = 0$ can be expressed as:

$$Q_0 = \frac{\theta W_0}{p_0}$$

Taking logs and first-differencing:

$$\ln Q_0 = \ln \theta + \ln W_0 - \ln p_0$$

$$\Delta \ln Q = \Delta \ln \theta + \Delta \ln W - \Delta \ln p$$

Using a first-order Taylor approximation, $\Delta \ln W \approx \frac{W_0 - W_{-1}}{W_{-1}} = \frac{p_{-1} Q_{-1}}{W_{-1}} \times \frac{\Delta p}{p_{-1}} + \frac{F_0}{W_{-1}} = \theta \Delta \ln p + \frac{F_0}{W_{-1}}$.

Noting that $\Delta \ln \theta = 0$ and substituting for $\Delta \ln W$:

$$\Delta \ln Q = \theta \Delta \ln p + \frac{F_0}{W_{-1}} - \Delta \ln p$$

$$\Delta \ln Q = (\theta - 1) \Delta \ln p + \frac{F_0}{W_{-1}}$$

Finally, computing signed elasticity as:

$$\xi = -\frac{\partial \Delta \ln Q}{\partial \Delta \ln p} = 1 - \theta.$$

gives the derived expression. □

This shows that as θ increases, the market becomes more inelastic (less elastic), and the price impact $\frac{\theta}{1-\theta}$ increases — every additional dollar of flows or borrowing has a larger impact on equity prices. Our assumed choice of $\theta = 0.8$ implies $\xi = 0.2$.

As an aside, note that when $\theta = 0$, $p_0 = p_{-1}$. However, this does not represent a Lucas-style asset pricing model because i) price does not evolve from $t = -1$ to $t = 0$ despite discount rates changing, and ii) elasticity is $\xi = 1$ which does not represent the perfectly elastic demand implied in the Lucas model ($\xi = \infty$). We discuss how the Lucas model can be derived as a special case of our model in Appendix F.4, where the proof relies on the ability of the financial intermediary to re-optimize and choose θ at $t = 0$, taking into account flows from the wealthy household.

3.3. Wealth Inequality

We first define the wealth of households and define our preferred measure of wealth inequality:

DEFINITION 1. *The period t wealth, for $t \in \{-1, 0\}$, is defined as:*

- *Wealthy Household:* $W_t^R \equiv p_t Q_t + B_t$,
- *Non-Wealthy Household:* $W_t^P \equiv \mathcal{O}^P - B_t$, $\mathcal{O}^P > B_t$.

The period t wealth share, for $t \in \{-1, 0\}$, is defined as:

- *Wealthy Household:* $\eta_t^R \equiv \frac{W_t^R}{W_t^R + W_t^P}$,
- *Non-Wealthy Household:* $\eta_t^P \equiv \frac{W_t^P}{W_t^R + W_t^P}$.

A few points are in order here. First, why do we consider the wealth position in periods $t \in \{-1, 0\}$ only? These are the wealth positions of relevance to understand the impact of inelastic markets — if we consider the wealth position at $t = 1$, it would trivially equal the income of each household

type. However, the evolution of the wealth share from $t = -1$ to $t = 0$ depends on the evolution of equity price, which is impacted by inelastic markets.

Second, we assume that the non-wealthy household owns non-marketable assets worth $\mathcal{O}^P$. These might consist of houses being used as primary residence or other illiquid assets which cannot be freely disposed. We make this assumption (along with constraining borrowing to be lower than the value of these assets) to ensure that non-wealthy households do not die in debt or have negative wealth positions (and negative wealth shares).

What, then, is the impact of equity price rises on wealth inequality? Consider the following proposition:

PROPOSITION 3. *The wealthy household share is increasing in equity prices:*

$$(3) \quad \frac{\partial(\eta_0^R - \eta_{-1}^R)}{\partial p_0} < 0$$

PROOF. Substitute for the definitions of wealth shares and take the first-order condition with respect to p_0 . This yields:

$$\frac{\partial(\eta_0^R - \eta_{-1}^R)}{\partial p_0} = \frac{Q_0(\mathcal{O}^P - B_0)}{(p_0 Q_0 + \mathcal{O}^P)^2}$$

which is positive when $\mathcal{O}^P > B_0$, hence true by assumption. $\square$

Therefore, by invoking the pricing equation, the wealthy household wealth share goes up whenever flows or borrowing is higher, θ is higher, Q_0 is lower, or fundamental price p_{-1} is higher. This establishes a novel link between wealth inequality and the inelasticity of markets.

3.4. Comparative Statics on Income

We turn to studying the effect of changing wealthy and non-wealthy income (endowment) on equity prices and wealth inequality. Our aim is to study the relationship between income inequality and wealth inequality. As it is no longer possible to derive closed-form analytical solutions for prices with respect to income shares, we rely on numerical simulations.

We choose the following parameter values to run the simulation: log utility, Dividend $D_1 = 2$,

elasticity parameter / share of equity in intermediary portfolio $\theta = 0.8$, discount factor $\beta = 0.99$, pre-period financial intermediary wealth $W_{-1} = 5$ and equity supply $\bar{Q} = 10$. We calibrate endowments to maintain a flat top 10% income share of 80%, with $\{e_0^P, e_0^R\} = \{2, 8\}$ and $\{e_1^P, e_1^R\} = \{8, 12\}$.

We first define our measure of income inequality:

DEFINITION 2. *The period t income, for $t \in \{0, 1\}$, is defined as:*

- *Wealthy Household:* $I_t^R \equiv e_t^R + \mathbb{I}_{t=1} D_1 Q_t$
- *Non-Wealthy Household:* $I_t^P \equiv e_t^P$

The period t income share, for $t \in \{0, 1\}$, is defined as:

- *Wealthy Household:* $\mathcal{I}_t^R \equiv \frac{I_t^R}{I_t^R + I_t^P}$
- *Non-Wealthy Household:* $\mathcal{I}_t^P \equiv \frac{I_t^P}{I_t^R + I_t^P}$

We study the impact on prices of varying individual incomes. We summarize these in the following:

PROPOSITION 4. *The partial derivatives of equity price with respect to endowments are:*

- *If $t = 0$ endowment of non-wealthy household increases, their demand for borrowing falls, hence prices decrease.*

$$\frac{\partial p_0}{\partial e_0^P} < 0$$

- *If $t = 1$ endowment of non-wealthy household increases, their demand for borrowing increases due to higher ability to repay debt, hence prices increase.*

$$\frac{\partial p_0}{\partial e_1^P} > 0$$

- *If $t = 0$ endowment of wealthy household increases, their demand for savings increases, hence flows increase and prices increase.*

$$\frac{\partial p_0}{\partial e_0^R} > 0$$

- *If $t = 1$ endowment of wealthy household increases, their demand for savings falls, hence flows fall and*

prices decrease.

$$\frac{\partial p_0}{\partial e_1^R} < 0$$

Given these results, we need to study how changing relative endowments effects prices. We find the following relationships, demonstrating that the effect of changing wealthy endowments dominates the opposite effect of changing non-wealthy endowments:

PROPOSITION 5. *The distribution of income matters for equity prices. Prices increase when current period wealthy income share increases.*

$$\frac{\partial p_0}{\partial \mathcal{I}_0^R} > 0$$

Finally, let us explore the relationship between income inequality and wealth inequality over time. As discussed above, endowments are calibrated such that $\mathcal{I}_0^R = \mathcal{I}_1^R = 0.8$. Under this calibration, we find that the top 10% wealth share *increases* from 0.625 to 0.644, a 3% increase. When we calibrate endowments such that top 10% income share falls from 0.8 to 0.75, top 10% wealth share *increases* from 0.625 to 0.7, a 12% increase. Intuitively, this happens because both the equity investment and borrowing channels kick in — a decrease in relative income for wealthy households prompts higher savings / flows into the financial intermediary, and an increase in relative income for non-wealthy household also prompts higher borrowing from the financial intermediary. Under inelastic markets, equilibrium forces kick in to ensure that the target portfolio allocation is maintained — this results in large equity price increases, which exacerbates wealth inequality.

3.5. Correlation between Wealth Inequality and Elasticity

Here, we re-establish the definition of wealth inequality for the quantitative model. To derive an empirically testable and tractable prediction, we consider a special case of the model without middle class households and housing assets. Under this scenario, the top 10% wealth share in the economy is given by:

$$(4) \quad \eta_t^R = \frac{W_t^R}{W_t^R + W_t^P} = \frac{p_t Q_t + B_t}{p_t Q_t}.$$

where we drop the assumption of other illiquid assets held by non-wealthy households for simplicity.

In the three-period model, we established the relationship between equity price and wealth inequality. We now derive the relationship between elasticity and wealth inequality:

PROPOSITION 6. *The change in top 10% wealth inequality is negatively related with the change in the inverse of the price multiplier. As markets become more inelastic between t and $t + 1$, the inverse of the price multiplier decreases, implying a negative difference.*

$$(5) \quad \eta_{t+1}^R - \eta_t^R = \underbrace{\frac{1 - \theta_{t+1}}{\theta_{t+1}} - \frac{1 - \theta_t}{\theta_t}}_{<0}$$

PROOF. Writing $p_t Q_t = \theta_t \tilde{W}_t$ and $B_t = (1 - \theta_t) \tilde{W}_t$, we get:

$$\eta_{t+1}^R - \eta_t^R = \frac{p_{t+1} Q_{t+1} + B_{t+1}}{p_{t+1} Q_{t+1}} - \frac{p_t Q_t + B_t}{p_t Q_t} = 1 + \frac{B_{t+1}}{p_{t+1} Q_{t+1}} - 1 - \frac{B_t}{p_t Q_t} = \frac{1 - \theta_{t+1}}{\theta_{t+1}} - \frac{1 - \theta_t}{\theta_t}$$

□

Intuitively, as markets become more inelastic, the price multiplier is larger, resulting in larger equity price changes and consequently higher increases in top 10% wealth positions (and shares).

This proposition offers us an empirically testable prediction of the model — the change in top 10% wealth share and the change in the value of bond holdings relative to equity holdings of inelastic investors ($\Delta \frac{B}{pQ}$) should be negatively correlated. This is indeed verified in the data, as we observed in the empirical motivation of the paper (Fig. 2). Our model therefore generates the result that as the change in the inverse of the price multiplier falls (and markets become more inelastic), the top 10% wealth share increases, as seen in the data.

4. Impulse Response Functions

What are the directional effects of income share shocks? Here, we examine how macro-variables evolve in the presence of inelastic markets under the broader trends in labour shares observed over the past few decades. The past three decades have witness increasing top 10% labour income shares, at the expense of falling labour income shares of the rest of the 90%. An active literature has sought to establish micro-foundations for the drivers of labour share divergence. Several factors

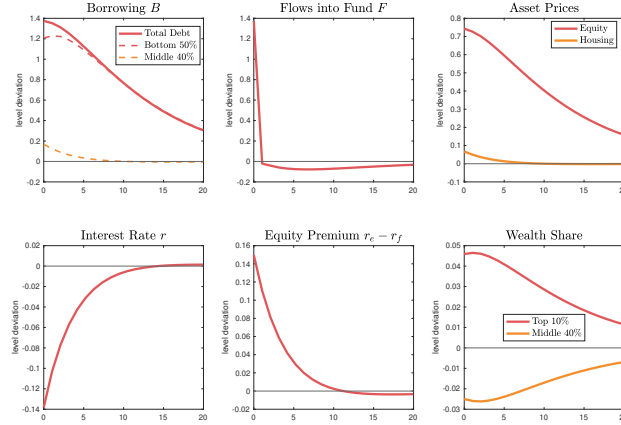
can drive divergence in income between the rich and the poor, including returns to education (Katz and Murphy 1992), automation (Acemoglu and Restrepo 2020), globalization (Autor et al. 2013), declining market power of workers (Stansbury and Summers 2020), rising market power of firms (De Loecker et al. 2020), and increasing profit share (Barkai 2020).

Our experiment considers three scenarios. In the first scenario, the wealthy household gets a positive income share shock which comprises of correlated positive shocks to labor income (endowment) and dividend income. In the second scenario, non-wealthy households get a negative labor income (endowment) shock. In the third scenario, middle class households get a negative labor income (endowment) shock, correlated with a positive housing dividend shock. In the third scenario, we allow for the borrowing constraint on middle class households to be occasionally binding. The results of the experiments are plotted in Figure 4.

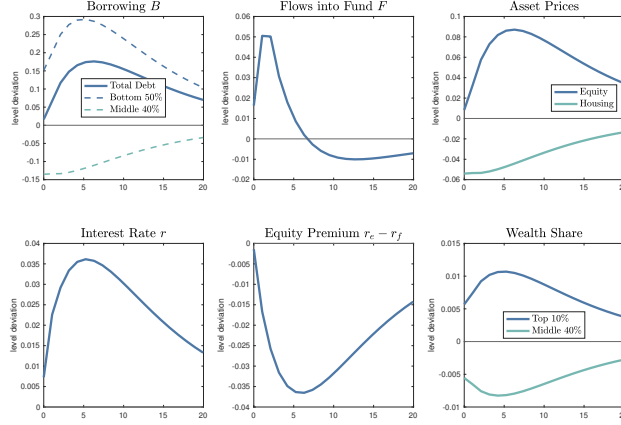
Positive Top 10% Income Shock. In Figure 4A, we simulate a simultaneous positive MIT shock to the labor share for the top 10% and a positive MIT shock to equity dividends. The results indicate that the wealthy want to save more — flows into the financial intermediary increase, which leads to increased bond holdings. Both non-wealthy and middle class households increase borrowing through the savings market, though non-wealthy households increase their borrowing more. The increased savings by the rich lead to a decline in interest rates. Due to inelastic markets, equity prices rise persistently above their steady-state levels. As a consequence of rates falling and higher dividends, the equity premium rises. As middle class households' borrowing constraint binds, they demand more housing, pushing up house prices slightly. The revaluation of the top 10% portfolio drives an increase in the wealth share of the top 10%. The increase is large enough to counteract the slight increase in middle class portfolio values — their wealth share (along with bottom 50% wealth share) drops. The results show the *equity investment channel* in action.

Negative Bottom 50% Income Shock. In Figure 4B, we examine a negative MIT shock to the labor share for the bottom 50%. Here, borrowing by the bottom 50% rises, part of which is provided by more savings from the middle class household. Therefore, net borrowing of the financial intermediary rises modestly upon impact, but steadily increases over time. The top 10% are incentivised to increase their flows into the intermediary, while middle class households are incentivised to save

A. Positive labour share shock for top 10%



B. Negative labour share shock to bottom 50%



C. Negative labour share shock to middle 50-90%

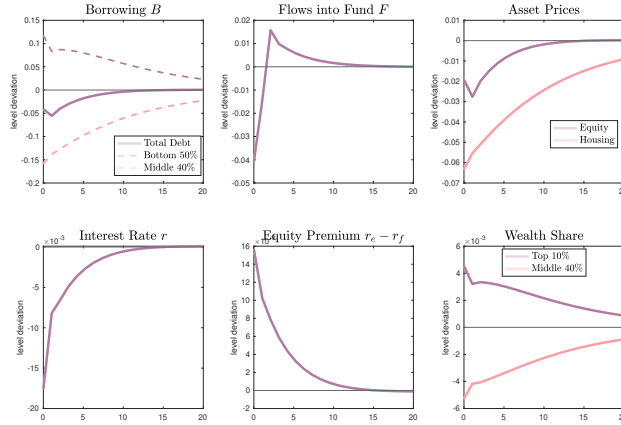


FIGURE 4. Panel A shows the IRFs in response to a positive labour share shock to the top 10%. Panel B shows IRFs in response to a negative labour share shock to the bottom 50%. Panel C shows IRFs in response to a negative labour share shock to the middle class, 50-90%.

instead of borrow in the risk-free bond due to the rise in interest rates (due to higher borrowing demand from non-wealthy). Similar to the positive shock, inelastic markets cause equity prices to rise persistently above steady-state levels. The combination of increased borrowing by the bottom 50% and the revaluation of the top 10% portfolio results in a higher wealth share for the top 10%. The middle class sells part of their housing to transfer wealth to the bond, which results in lower house prices and lower middle class wealth share. These results show the *borrowing channel* in action.

Negative Middle 50-90% Income Shock. In Figure 4C, we simulate a simultaneous negative MIT shock to the labor share for the middle class and a positive MIT shock to housing dividends (rental income). On impact, middle income households reduce their demand for housing, which reduces house prices. As the borrowing constraint continues to bind, this translates into lower borrowing by the middle class, which pushes down interest rates. This incentivises bottom 50% households to step in to borrow, which implies that total borrowings fall marginally. Reduced aggregate borrowings imply outflows from the financial intermediary, hence equity prices are lower. However, the fall in house prices is larger than the fall in equity prices, implying that the wealth share of the middle class reduces and remains persistently below the top 10% wealth share.

In summary, under inelastic markets, shocks to income inequality translates into wealth inequality through both the equity investment and the borrowing channel. The increase in savings by the top 10% and the increase in borrowing by middle class and non-wealthy households boosts equity valuations, and thereby increase the wealth of the wealthy due to portfolio heterogeneity.

4.1. IRFs under Different Elasticities

How much do the impulse response functions depend upon the level of inelasticity in the model, controlled by varying the parameters θ^* and χ^{MF} ? Intuitively, reducing χ^{MF} is equivalent to reducing inelasticity in the model. Similarly, increasing θ^* implies that the borrowing response will be dampened as demand for bonds will be lower. In Figures 5 and 6, we consider our variables of interest while varying θ^* and χ^{MF} . The results are qualitatively similar, which gives us confidence that our specific calibration choices are not driving our results. Nonetheless, there are some

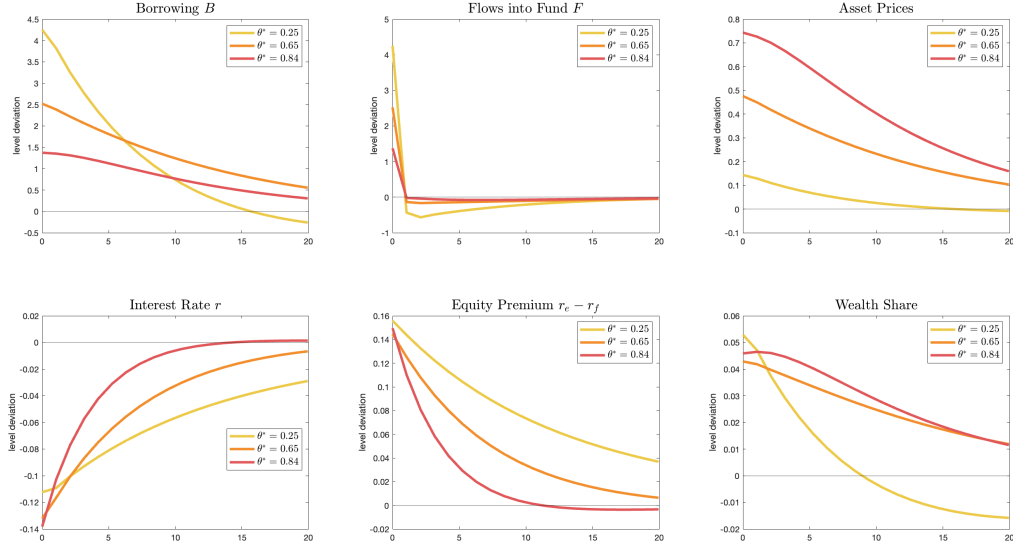
interesting outcomes which we discuss next.

In Figures 5A and 6A, we consider the effects of a positive top 10% labour share shock and negative bottom 50% labour share shock, while varying θ^* to take three values, 0.25, 0.65, and our calibrated case, 0.84. With lower target allocations, a positive wealthy labour share shock results in larger borrowing and flows responses. However, the impact on equity prices is lower, following from the pricing equation 15. The fall in interest rates (and increase in equity premia) are more persistent. The increase in wealth share also continues to be persistent, but the impact of the flows channel is reduced. With a negative bottom 50% labour share shock, the effects are much more pronounced and persistent. This is because with a lower θ^* , the borrowing channel is much more prominent, resulting in large increases in wealth shares driven both by more persistent (but smaller) asset price increases, and higher bond holdings.

In Figures 5B and 1A, we consider the effects of a positive top 10% labour share shock and negative bottom 50% labour share shock, while varying χ^{MF} to take three values, 1, 20, and our calibrated case, 171. When adjustment costs are low at $\chi^{MF} = 1$, it is unsurprising that borrowing and flow responses are smaller. However, perhaps counterintuitively, equity price increases are *higher* — the fund is able to adjust quickly to increases in equity premia, and their higher demand pushes up asset prices more. Most interestingly, in the case of the negative bottom 50% shock, lower adjustment costs result in *falling* equity prices. This is because the fund is agile enough to reduce their equity allocation in the face of lower equity premium, despite higher flows, which is instead allocated to higher borrowing. Quantitatively, the difference in effects when $\chi^{MF} = 20$ and $\chi^{MF} = 171$ is very small. Therefore, the response of endogenous variables to increasing χ^{MF} is a concave function.

A. Positive labour share shock for top 10% - Varying θ^*

Top 10% Endowment Shock Response, varying θ^*



B. Positive labour share shock for top 10% - Varying χ^{MF}

Top 10% Endowment Shock Response, varying χ^{MF}

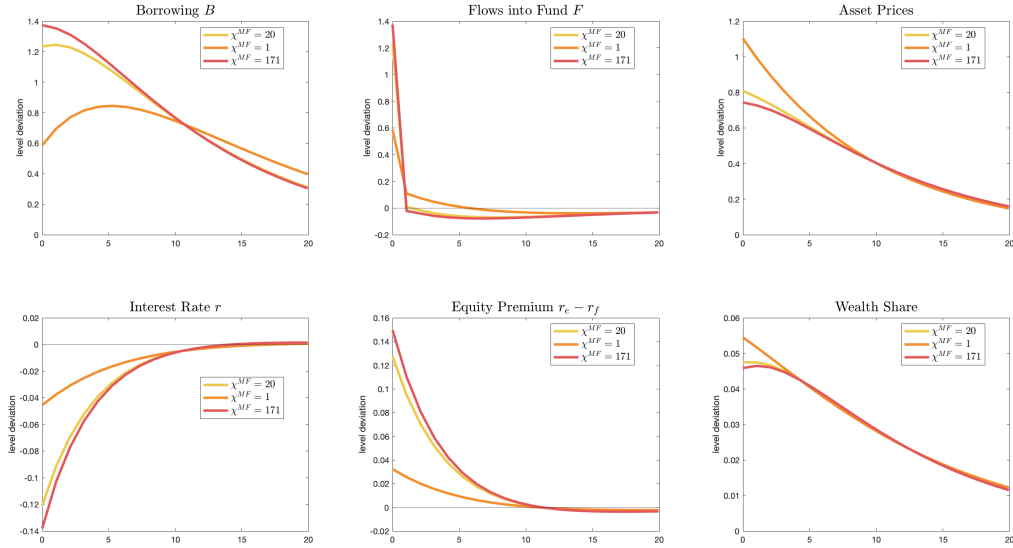
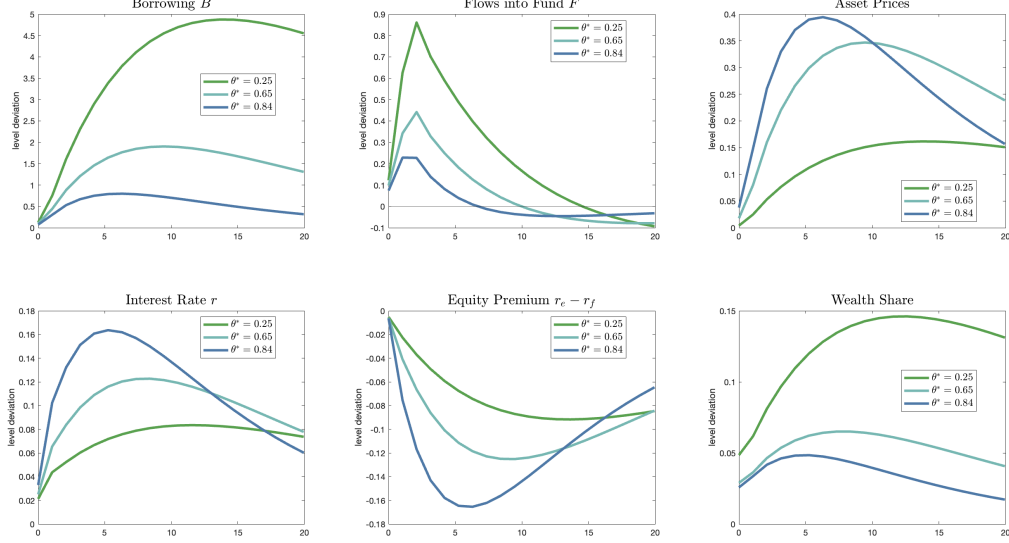


FIGURE 5. Both panels shows the IRFs in response to a positive labour share shock to the top 10%. Panel A shows how IRFs vary as θ^* varies. Panel B shows how IRFs vary as χ^{MF} varies.

A. Negative labour share shock for bottom 50% - Varying θ^*

Bottom 50% Endowment Shock Response, varying θ^*



B. Negative labour share shock for bottom 50% - Varying χ^{MF}

Bottom 50% Endowment Shock Response, varying χ^{MF}

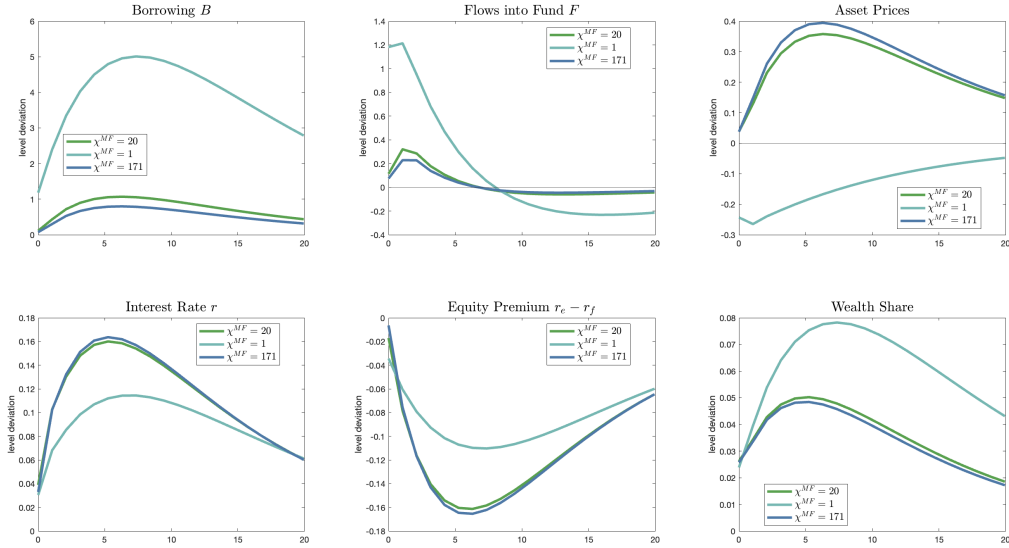


FIGURE 6. Both panels shows the IRFs in response to a positive labour share shock to the top 10%. Panel A shows how IRFs vary as θ^* varies. Panel B shows how IRFs vary as χ^{MF} varies.

4.2. Equity Prices: Inelastic vs. Elastic Markets: Full Model

Figure 7 plots the response of equity prices to labour share shocks and dividend shocks in a model with middle class households and housing.

Negative non-wealthy income shocks continue to not matter in the elastic markets model, while prices respond much more in the inelastic markets model compared to the case with no middle class households. Positive dividend shocks continue to have a muted impact in the inelastic markets model.

The biggest difference is in the response of prices to a positive wealthy income shock. The interest rate decrease is bigger in the model with middle class, which leads to a much larger price impact in the elastic markets model upon impact. This is because in this model, due to middle class households being borrowing constrained, higher equity investment from the rich cannot be fully accommodated by frictionlessly increasing borrowing. The capacity of the non-wealthy household to borrow more is also limited (their endowments are lower than in the case without middle class). Therefore, interest rates need to fall more to induce non-wealthy households to borrow.

However, the elastic markets model cannot produce high persistence. The effect on prices reverts to steady state in eight periods, while in the inelastic markets model, the effect remains persistent for more than twenty periods, implying that the overall impact on wealth inequality over time is larger under inelastic markets.

We thus conclude that equity price responses under inelastic and elastic markets are qualitatively and quantitatively different, resulting in very different outcomes for wealth inequality.

5. Simulation: Market Capitalization, Total Debt, Divergence in Inequality

In Section 4.2, we demonstrated that the model can simultaneously match rising equity prices, rising total debt, and increasing divergence between income and wealth inequality in response to income and dividend shocks. How does our model perform in matching these three trends in a longer horizon simulation, rather than in a one-time shock exercise?

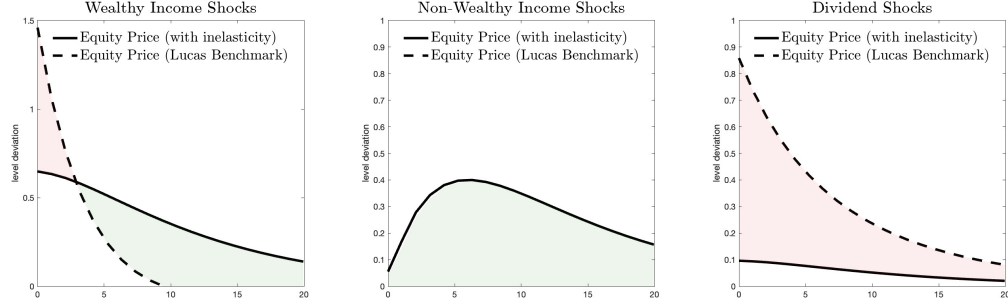


FIGURE 7. Panel A shows equity price IRF in response to a positive labour share shock to the top 10%. Panel B shows equity price IRF in response to a negative labour share shock to the bottom 50%. Panel C shows equity price IRF in response to a positive equity dividend shock.

In Figure 8, we plot the growth rate of market capitalization, the growth rate of total debt outstanding, and the differential (between top 10% and bottom 50%) of growth rates of income and wealth, under a perfect foresight simulation with expectation errors post-2008.

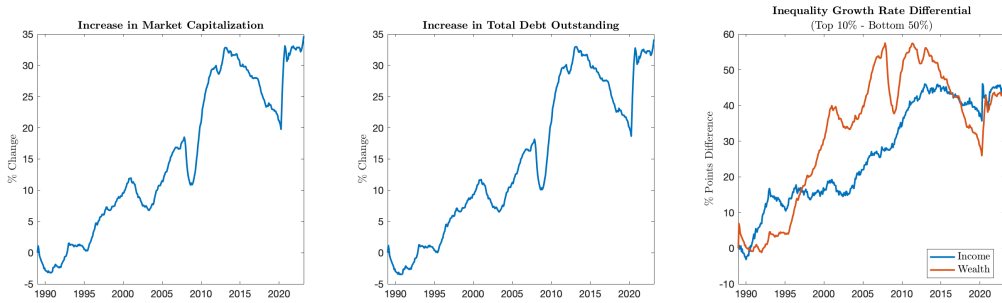


FIGURE 8. Dynamics of pQ , b and Inequality: Perfect Foresight with Expectation Errors post-2008

Note: The perfect foresight simulation initialises the economy at the January 1989 level of labour income shares and dividends. Then, agents are provided with the full realisation of income shares from January 1989 to December 2007. From January 2008, agents expect income shares to follow a smoothed trend. Each period, they make expectation ‘errors’ equivalent to the deviation of the actual income share relative to the trend. The figure plots percentage deviation of wealth shares relative to its 1989 level.

As expected from the IRF exercise, the model simulation does well in replicating the large increase in market capitalization as well as in total debt outstanding. In fact, the two series inherit each other’s dynamics given the simplified structure of our model. Crucially, the model is also able to capture the faster increase in the wealth growth rate differential relative to the income growth rate differential — the simulated version of the Inequality Multiplier.

However, the model does not perform well in quantitatively matching the magnitude of increase in market cap, debt, and wealth growth rate differential. This is unsurprising given that the model

focuses on the mechanism of inelastic markets at the cost of not considering other factors which drive equity market capitalization or debt, such as the endogenous decision of governments to issue more debt, company decision to list on equity markets, evolution of corporate profits etc. We conclude that the mechanism of inelastic markets delivers us qualitatively correct trends, and can quantitatively explain part of the magnitude of increase across the three measures.

6. Note: Elasticities

We review some concepts from Microeconomics and [Koijsen and Yogo \(2025\)](#). Suppose we had one fund with a mandate to hold θ of equities. Then their holdings of shares Q is

$$Q = \theta \frac{W}{P},$$

where W is the funds wealth and P is price of equity. Suppose there is a small movement in the price P . We can take logs

$$\log(Q) = \log(\theta W) - \log(P).$$

The price effect (Marshallian) effect says that *holding W constant*, a 1% rise in the price makes the fund want 1% fewer shares as the denominator P got bigger. The *uncompensated* Marshallian elasticity is

$$\left. \frac{\partial \log(Q)}{\partial \log(P)} \right|_W = -1.$$

This is also known as the *price effect*.

The *wealth effect* says that because of θ , a 1% price increase also raises the fund's wealth by θ %. The mandate reinvests that extra wealth into the stock

$$\frac{\partial \log(W)}{\partial \log(P)} = \theta.$$

The Hicksian elasticity is the sum of the two

$$(6) \quad \frac{\partial \log(Q)}{\partial \log(P)} = -1 + \theta.$$

So the unsigned Hicksian elasticity only reports the magnitudes whilst the signed gives the negative sign.

Signed Hicksian elasticity : $\varepsilon_{i,\text{signed}}^H = -(1 - \theta_i)$

Unsigned Hicksian elasticity : $|\varepsilon_i^H| = \zeta_i = 1 - \theta_i$